TX-OMICS FOLLOW-UP v3

Gene Set Breakdown

Shared transcriptomic responses to sleep–wake manipulations

Rosensweig, Shah et al. · bioRxiv 2026.02.28.708752

17
gene-set tiers

1162
genes (tier sum)

2,694
FlyBase stocks mapped

6
homeostatic core + 19 screen candidates

Prepared for Dr. Ravi Allada · July 2026

BIOLOGICAL FRAMING

Why these gene sets exist

Question

Is there a homeostatic “sleeper” gene battery—analogous to per for circadian timing—that tracks sleep–wake history across diverse manipulations?

Approach

Brain transcriptomics across 7 datasets: mechanical SD, thermo/optogenetic activation, THIP, and baseline cycling. Genes ranked by fold-change consistency (FC ≥ 0.5 or ≥ 1) and cross-dataset frequency.

This deck

17 tiered CSV gene lists in Breakdown/, each with per-set FlyBase stock workbooks. Sets move from broad rebound tiers → history correlates → curated mechanistic priorities.

ORGANIZATION

Six logical categories · 17 gene-set files

1

FC0.5 · Sleep rebound

2 tiers · 141 genes

Genes consistently up/down with sleep rebound ($log_2FC \geq 0.5$) across manipulations

2

FC0.5 · Wake rebound

5 tiers · 647 genes

Wake-enriched rebound responders; largest tiered collection

3

FC1 · Sleep / Wake

4 tiers · 114 genes

Stricter FC ≥ 1 filter — higher-confidence, smaller lists

4

Sleep history correlates

2 sets · 215 genes

Expression tracks prior sleep/wake history (± literature via fl.ai)

5

History × Rebound overlap

2 sets · 20 genes

Dual homeostatic signature: history correlate ∩ rebound responder

6

Curated priorities

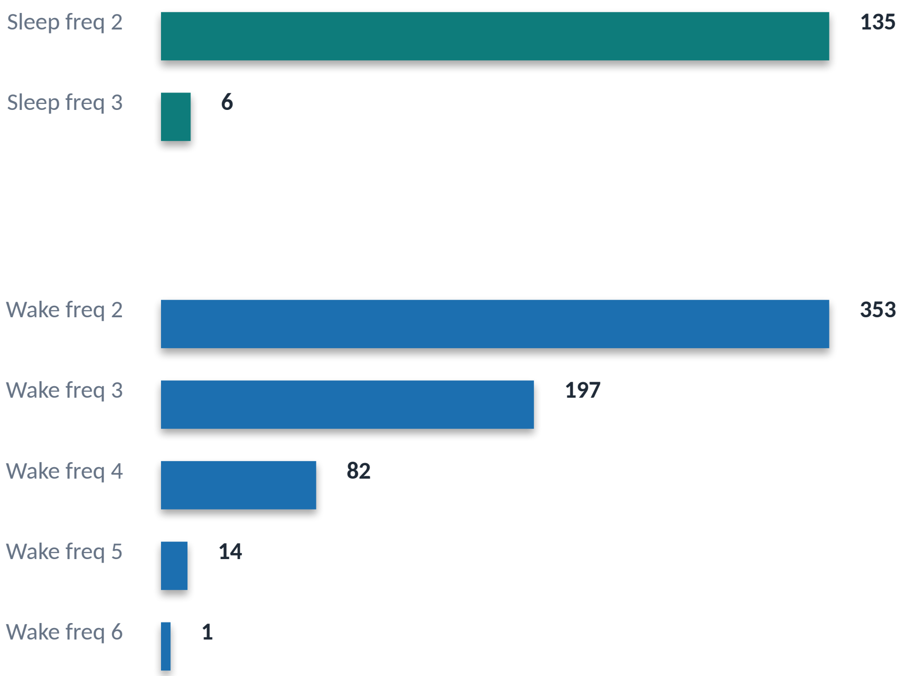
2 sets · 25 genes

6-gene homeostatic core + 19 mechanistic screen candidates (nsyb/elav)

Frequency tiers (freq 2–6): number of datasets in which a gene passes the FC threshold — higher freq = more robust.

Consistent sleep/wake responders by FC threshold & frequency

FC0.5 · genes per frequency tier



FC1 (stricter) · all tiers

Tier	Genes	Stocks
Wake · freq 2	77	156
Sleep · freq 2	16	28
Wake · freq 3	14	26
Wake · freq 4	7	8

Key pattern: Wake rebound sets dominate at FC0.5 (353 genes at freq 2 alone). Stricter FC1 collapses to 114 genes — useful for high-confidence follow-up.

Homeostatic signatures beyond raw rebound

215

history-correlate genes

Pos correlates (n=149): genes whose expression rises with prior wake.
Neg correlates (n=66): genes tracking prior sleep.

20

overlap genes (dual signature)

History(−) × Rebound(+) → 5 genes (e.g. CG9377)
History(+) × Rebound(−) → 15 genes (e.g. AstA-R2, Mip pathway)

Interpretation for mechanistic follow-up

Overlap sets enrich for genes with opposing history vs. rebound dynamics — prime homeostatic regulators.
fl.ai literature categories (sleep / circadian / both) annotate each gene; high-confidence overlaps feed the mechanistic screen list.
Per-gene CSVs include cycling status, manipulation correlations (MechSD, TrpA1, THIP), and PMC-backed rationales.

CURATED PRIORITIES

Homeostatic core + mechanistic screen candidates

6-gene homeostatic core

unc79 · SIFa · rumpel · AstA-R2 · Trhn · RpL23

NALCN channel auxiliaries, neuropeptide signaling, glial transport, serotonin biosynthesis, ribosome biogenesis — convergent homeostatic biology.

25

FlyBase stocks

Top mechanistic screen candidates (nsyb-GAL4 / elav-GAL4)

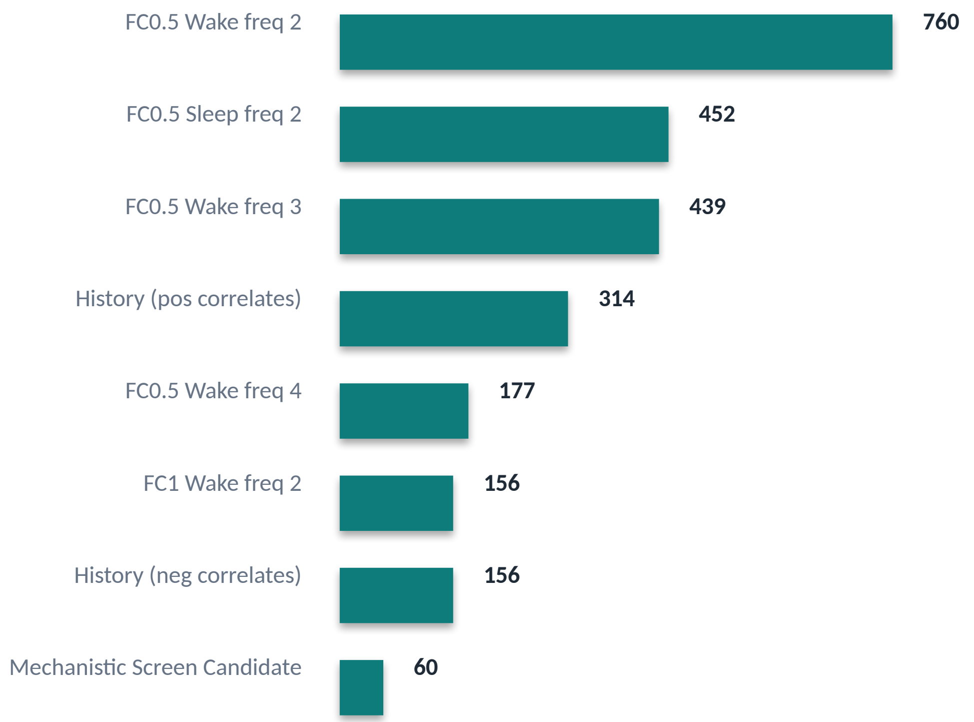
#	Gene	Mechanism	Priority
1	unc79	Homeostatic	**High**
2	SIFa	Homeostatic	**High**
3	rumpel	Homeostatic	**High**
4	AstA-R2	Homeostatic	**High**
5	Trhn	Homeostatic	**High**
6	RpL23	Homeostatic	**High**
7	na	Homeostatic	**High**
8	unc80	Homeostatic	**High**
9	BomBc2	Immune	**High**
10	Mip	Neuropeptide	**High**

Full ranked list: 19 genes · Tx-Omics_Mechanistic_Screen_Candidates_nsyb_elav.csv

REAGENT COVERAGE

FlyBase stocks per gene-set tier

Largest tiers by stock count



2,694

stocks across all tiers

2,504

alleles mapped

Each tier → 6 priority stock tables
(Bloomington / VDRC RNAi, alleles, non-stock-center)

Workbooks live in Breakdown/Per Gene Set Runs/ · summary in combination_counts_summary.csv

Takeaways

17 tiered gene lists span rebound consistency (FC0.5/FC1), sleep-history correlates, and dual-signature overlaps.
Frequency bins prioritize robust responders across 7 manipulation datasets — wake tiers are largest; FC1 narrows to high-confidence targets.
6-gene homeostatic core + 19 ranked mechanistic candidates are ready for nsyb/elav RNAi screening with mapped FlyBase reagents.
Next: prioritize overlap + high-freq FC1 wake genes for behavioral validation; stock tables are generated per tier.

data/gene_sets/Tx-Omics-FollowUp_v3/Breakdown/